

(a) Complete the table by writing the contents of the ACC after the execution of each instruction.

| Current contents of the ACC | Instruction | Contents of the ACC after the execution of the instruction |
|-----------------------------|-------------|------------------------------------------------------------|
| 0000 1111                   | AND 101     |                                                            |
| 0000 0000                   | LDM #100    |                                                            |
| 0000 0001                   | XOR &F1     |                                                            |
| 0001 0001                   | CMP 101     |                                                            |

[4]

(b) The Von Neumann model for a computer system uses registers.

Describe the role of the Memory Address Register (MAR) and Memory Data Register (MDR) in the fetch-execute (F-E) cycle.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(c) Assembly language instructions are grouped.

Complete each statement by writing the name of the appropriate instruction group.

Loading data into the accumulator is an example of an instruction in the

..... group. Incrementing the index register is an example

of an instruction in the ..... group. Branching to another

address is an example of an instruction in the ..... group.

[3]

